

ASITHA ISURU

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PROFESSIONAL SUMMARY

Results-oriented Industrial IT undergraduate seeking a Software Engineering Internship to apply full-stack development expertise to solve complex challenges.. Proven ability to design, build, and deploy practical applications, from management systems to a sophisticated computer vision project for process optimization. Skilled in Java, PHP, and the MERN stack, with a solid foundation in databases, REST APIs, and Agile practices. Eager to leverage technical skills in a collaborative team environment to contribute to innovative software solutions.

EDUCATION

B.Sc. (Hons) in Industrial Information Technology, Uva Wellassa University of Sri Lanka

Expected 2026 | GPA : 3.54

Relevant Coursework: Object-Oriented Programming, Software Engineering, Database Management Systems, Web Application Development, Computer Networks, Operating Systems, Data Structures and Algorithms, Software Quality Assurance, Human-Computer Interaction, IT Project Management, Mobile Computing

WORK EXPERIENCE

Customer Support Technician | Dialog Axiata (PVT) LTD, Colombo

December 2021 – May 2022 | Full-time

- Gained hands-on experience in IT support and problem-solving by diagnosing and resolving systems, hardware, and network connectivity issues for internal staff, minimizing workflow disruptions.
- Strengthened technical deployment skills by setting up and configuring employee devices, ensuring a seamless onboarding process.
- Developed critical communication and prioritization skills by managing daily IT support tickets and escalating complex technical issues to specialized teams to maintain service-level agreements.

CERTIFICATES

Oracle Certified Professional | Oracle University | 2025 May
Github Foundations | GitHub | 2025 January
Google Workspace Essentials | Udemy | 2021 October
CSS and JavaScript Crash Course | Udemy | 2021 October
Adobe Photoshop Ultimate Course | Udemy | 2021 October
Machine Learning with Python - Beginner | Udemy | 2021 October
Microsoft 365 Ultimate Course | Udemy | 2021 October
Get Connected | Cisco Networking Academy | 2021 July
Introduction to IoT | Cisco Networking Academy | 2021 July
Introduction to Packet Tracer | Cisco Networking Academy | 2021 July
NDG Linux Unhatched | Cisco Networking Academy | 2021 July
Diploma Course in English | Yale College, Miriswatta | 2019

KEY PROJECTS

Final Year Project - Research (ongoing) | Uva Wellassa University

Technologies : MongoDB, Express, React, Node, OpenCV, OpenMVG, OpenMVS, PyTorch

- Developing a full-stack, computer vision-driven decision support system to optimize quartz gemstone cutting using 3D reconstruction, computational geometry, and AI-based analysis.

Impact & Performance Metrics

- Built a fully automated photogrammetry pipeline achieving ~30-minute average 3D reconstruction time using GPU acceleration with CUDA integration.
- Processed 30+ quartz gemstones (~120 video samples) to generate accurate 3D models
- Achieved <2-3% volume error compared to physical measurements (water displacement validation)
- Developed a custom yield optimization engine capable of generating cutting strategies in ~15-20 minutes
- Demonstrated ~12% yield improvement over manual cutting estimates in preliminary testing
- Designed a system targeting 15–20% material waste reduction, addressing industry loss rates of ~30–35%
- Engineered a low-cost alternative to \$100K+ industrial scanning systems using consumer-grade imaging and open-source tools

GitHub Repo : <https://github.com/asithaisuru/quartz-optimizer>

Lan-Play (ongoing) | Self-Directed Project

Technologies : MongoDB, Express.js, React, Node.js

- A collaborative, network-hosted music platform where users can add and manage shared playlists using YouTube links in real-time.

Impact & Performance Metrics

- Implemented WebSocket-based real-time synchronization with a low-latency communication system enabling instant playlist updates across connected users and seamless client-side state synchronization.
- Built a multi-user collaborative environment supporting shared playlist management
- Introduced a unique audio announcement feature, allowing users to attach personalized messages that are played at the start of songs

GitHub Repo : <https://github.com/asithaisuru/lan-play>

Class Management System | Oasis Higher Educational Institute, Bemmulla

Technologies : MongoDB, Express, React, Node

- A web-based system for managing student academic activities, including attendance, grading, and fee tracking.

Impact & Performance Metrics

- Designed a MongoDB-based data architecture with modular collections and indexed queries, enabling the system to scale efficiently for 100–200+ student records with future horizontal scaling capability
- Reduced attendance recording time by ~60–70% compared to manual methods
- Improved data accuracy by minimizing human entry errors through structured digital input
- Enabled instant retrieval of student data, reducing lookup time from minutes to seconds
- Centralized academic operations, eliminating paper-based record redundancy

GitHub Repo : https://github.com/asithaisuru/project_oasis

REFERENCES

B.K. Sineth Sankalpa

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